

Homework 7

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MATH 421 — The Theory of Single Variable Calculus
Section 003
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Collaboration Statement

Problem 1 (*Ross Chapter 2 11.4*)

Repeat Exercise 11.2 for the sequences:

$$w_n = -(2)^n, x_n = 5^{(-1)^n}, y_n = 1 + (-1)^n, z_n = n \cos\left(\frac{n\pi}{4}\right).$$

Question 11.2 asks you to do the following for each sequence:

- (a) For each sequence, give an example of a monotone subsequence.
- (b) For each sequence, give its set of subsequential limits.
- (c) For each sequence, give its $\limsup$ and $\liminf$.
- (d) Which of the sequences converges? diverges to $+\infty$? diverges to $-\infty$?
- (e) Which of the sequences is bounded?

Discussion.

Proof.

□

Problem 2 (*Ross Chapter 2 11.8*)

Use Definition 10.6 and Exercise 5.4 to prove $\liminf s_n = -\limsup(-s_n)$ for every sequence (s_n) .

Discussion.**10.6 Definition:**

Let (s_n) be a sequence in $\mathbb{R}$. We define:

$$\limsup s_n = \lim_{N \rightarrow \infty} \sup\{s_n : n \geq N\}$$

and

$$\liminf s_n = \lim_{N \rightarrow \infty} \inf\{s_n : n \geq N\}$$

5.4 Exercise:

Let S be a nonempty subset $\mathbb{R}$, and let $-S = -s : s \in S$. Prove $\inf S = -\sup(-S)$. *Hint: For the case $-\infty < \inf S$, simply state that this was proved in Exercise 4.9.*

Proof.

□

Problem 3 (Ross Chapter 2 11.10)

Let (s_n) be the sequence of numbers in Fig. 11.2 listed in the indicated order.

- Find the set S of subsequential limits of (s_n) .
- Determine $\limsup s_n$ and $\liminf s_n$.

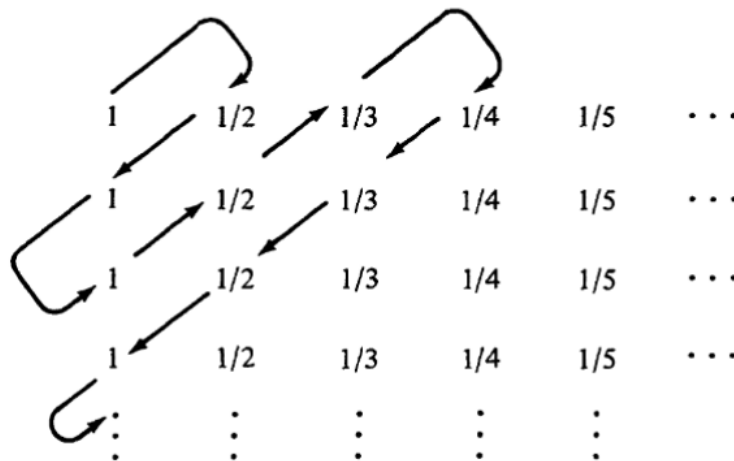


FIGURE 11.2

Discussion.

Proof.

□

Problem 4 (*Ross Chapter 2 12.2*)

Prove $\limsup |s_n| = 0$ if and only if $\lim s_n = 0$. **Discussion.**

Proof.

□

Problem 5 (*Ross Chapter 2 12.3*)

Let (s_n) and (t_n) be the following sequences that repeat in cycles of four:

$$(s_n) = (0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, \dots)$$

$$(t_n) = (2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, \dots)$$

Find

- | | |
|---------------------------------|---------------------------------|
| (a) $\liminf s_n + \liminf t_n$ | (e) $\limsup s_n + \limsup t_n$ |
| (b) $\liminf(s_n + t_n)$ | (f) $\liminf(s_n t_n)$ |
| (c) $\liminf s_n + \limsup t_n$ | (g) $\limsup(s_n t_n)$ |
| (d) $\limsup(s_n + t_n)$ | |

Discussion.

Proof.

□

Problem 6 (*Ross Chapter 2 12.10*)

Prove (s_n) is bounded if and only if $\limsup |s_n| < +\infty$.

Discussion.

Proof.

□